EGYPTIAN RUSSIAN UNIVERSITY

AI Recruitment & Resume Screening

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Problem Statement

Addressing the Challenges in Recruitment Screening

Slow Process

Manual resume screening is **time-consuming** and repetitive, leading to delays in the hiring process and affecting overall productivity.

Inconsistent Formats

Differing CV formats and wording across candidates create complications in evaluation, making it difficult for recruiters to compare qualifications effectively.

Semantic Gaps

Keyword screening often overlooks **semantic meaning**, resulting in potential misjudgments about candidates' skills and experiences relevant to job roles.

Project Objectives

01

The system will **upload and parse** various CV formats efficiently.

02

It will **rank candidates** by comparing CVs to job descriptions.

03

The tool will **detect biased wording** in job descriptions for fairness.

04

It will **generate synthetic CVs** for demonstration and testing purposes.

05

The system will **transform resume writing styles** for improved presentation.

System Workflow Overview

Connecting AI Modules for Recruitment Efficiency

This system integrates multiple AI modules into a cohesive workflow, facilitating seamless CV uploads, parsing, ranking, bias detection, synthetic CV generation, and interactive querying through the RAG assistant interface.

System Architecture

The system architecture encompasses several key components:

- **Streamlit UI:** Serves as the central interface connecting all modules.
- **Parsing Service:** Handles CV uploads and data extraction.
- **Ranking Service:** Compares job descriptions with CVs using AI techniques.
- **Bias Detection Service:** Analyzes job wording for fairness.
- **Synthetic CV Generation:** Creates demo CVs for testing.
- **Style Transformation Module:** Rewrites resumes without altering factual content.
- **RAG Assistant:** Facilitates question answering based on CV context.
- **Storage:** Manages local data and processed outputs.

Dataset Overview

Key Data Insights

Unique CVs

The project utilizes **20 unique Data Engineer CVs** for testing purposes, showcasing a variety of skills and experiences relevant to the data engineering field to ensure comprehensive evaluation.

Structured Records

Each CV is meticulously parsed into **structured records**, enabling efficient analysis and comparison against job descriptions, while maintaining clarity and accessibility of the data for further processing.

Privacy Assurance

We prioritize **data privacy**; therefore, private CV details are not disclosed in documentation or screenshots, ensuring that sensitive information remains protected throughout the demonstration and testing phases.

Resume Ranking Module

Job Description Matching

This module compares user-submitted job descriptions with uploaded CVs, utilizing OpenAI embeddings for accurate ranking, ensuring a comprehensive analysis of skills and qualifications.

Bias Detection Module

Ensuring Fair Recruitment

The Bias Detection Module analyzes job descriptions for biased language, suggesting neutral alternatives to promote fairness, inclusivity, and a diverse candidate pool throughout the recruitment process.

Synthetic CV Generation

Simulated CV Data

This module generates realistic synthetic CVs for demonstration purposes, offering structured records that simulate candidate profiles without revealing any private information.

Resume Style Transformation

Enhancing resume presentation effectively

The system allows users to upload resumes and transforms them into **professional, ATS-friendly**, or concise styles, improving wording while preserving original facts without altering experience or qualifications.

RAG Assistant Overview

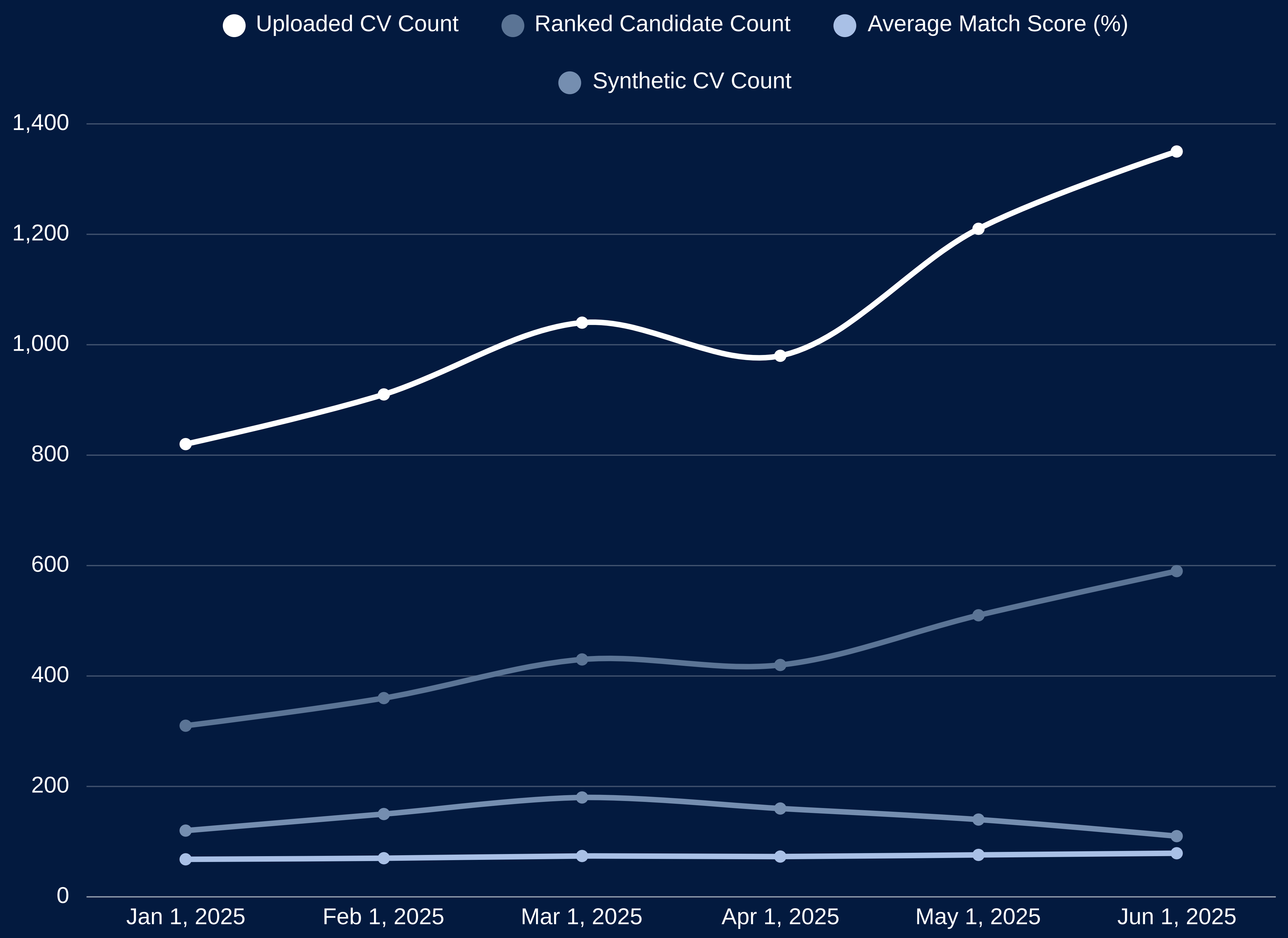
Understanding Context Retrieval

The RAG Assistant enhances recruitment by allowing users to ask contextual questions based on CV data, ensuring accurate responses grounded in relevant information.

Responsible AI Decision Support

The system supports recruiters with ranking scores, bias checks, RAG-based evidence, and dashboard metrics. It helps organize review, but final hiring decisions remain the responsibility of human reviewers.

Human-centered recruitment review with explainable AI outputs



Source: AI-generated data. Replace or verify before use.

Results and Testing Summary

01

The system processed 20 CV records successfully without failures.

02

It generated 35 synthetic CVs for testing and demonstration.

03

Average match score across candidates was 52.8%, indicating relevance.

04

The system's functionality was validated through a 200 OK response.

What's Next for AI?

Exploring future advancements in recruitment technology together

THANK YOU FOR YOUR TIME AND FOR LISTENING.